TERTIARY INFOTECH ACADEMY

LAB 06
Prove Backup and Restore Readiness

AWS Certified SysOps Administrator Associate Training
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Scenario
Northstar Retail reports successful backup jobs but has never proven that the protected data can be restored within its business objectives.

Objective
Apply K4, A1, A4 and A6 by protecting an S3/EBS-style workload, restoring into isolation and measuring actual recovery evidence.

Deliverable
A backup plan, selection, restore checklist, recovery evidence and measured RTO/RPO report.

Services and tools
AWS Backup · S3 · EBS · KMS · CloudWatch Events

Safety and cost boundary
- Use only the trainer-authorised learner account and Region.
- Replace placeholders and tag learner-owned resources.
- Capture evidence before cleanup.
- Verify deletion and report retained chargeable resources.

Detailed procedure

Step 1 — Read `RECOVERY-OBJECTIVES.md`; agree the workload scope, RTO, RPO, retention and restore-validation method.
```
open RECOVERY-OBJECTIVES.md
```

Evidence note: _____

Step 2 — Validate `backup-plan.json` and replace the learner-specific vault and role values.
```
python3 -m json.tool backup-plan.json
```

Evidence note: _____

Step 3 — Create the backup vault and plan with the trainer-approved encryption key and retention.
```
aws backup create-backup-vault --backup-vault-name SysOps-Lab06-REPLACE_SUFFIX
aws backup create-backup-plan --backup-plan file://backup-plan.json | tee evidence/backup-plan.json
```

Evidence note: _____

Step 4 — Create the resource selection from `backup-selection.json`; target only resources tagged for this course.
```
aws backup create-backup-selection --backup-plan-id REPLACE_PLAN_ID --backup-selection file://backup-selection.json | tee evidence/selection.json
```

Evidence note: _____

Step 5 — Start an on-demand backup and record start/completion times, recovery-point ARN and status.

```
aws backup start-backup-job --backup-vault-name SysOps-Lab06-REPLACE_SUFFIX --resource-arn
REPLACE_RESOURCE_ARN --iam-role-arn REPLACE_BACKUP_ROLE_ARN | tee evidence/backup-job.json
```

Evidence note: _____

Step 6 — Restore to an isolated learner-owned target; never overwrite the original resource.

```
aws backup start-restore-job --recovery-point-arn REPLACE_RECOVERY_POINT_ARN --iam-role-arn
REPLACE_RESTORE_ROLE_ARN --metadata file://restore-metadata.json | tee evidence/restore-
job.json
```

Evidence note: _____

Step 7 — Validate restored object count, checksum or application-readable state; record the exact method and result.

```
shasum -a 256 sample-data.txt | tee evidence/source-checksum.txt
```

Evidence note: _____

Step 8 — Complete `RECOVERY-REPORT.md` with actual RTO, observed RPO, integrity result, gaps and owners.

```
open RECOVERY-REPORT.md
```

Evidence note: _____

Step 9 — Capture backup and restore job status as JPEG evidence and compare with the architecture reference.

```
open EXPECTED-ARCHITECTURE.jpg
```

Evidence note: _____

Step 10 — Run validation; remove the restored target and learner-created plan after retention/permission checks.

```
bash validate.sh
bash cleanup.sh | tee evidence/cleanup.txt
```

Evidence note: _____

Verification

A completed recovery point is restored to an isolated target, integrity is checked, actual recovery is compared with RTO/RPO, gaps have owners, and validation emits LAB 06 PASS.

Acceptance criteria

- Scope uses course tags
- Restore target is isolated
- Retention is explicit
- Integrity validation is objective
- Actual RTO/RPO and gaps are recorded

Evidence checklist

□ Backup plan and selection

□ Backup job

□ Restore job

□ checksum evidence

□ completed RECOVERY-REPORT.md

□ JPEG evidence

□ cleanup.txt

Troubleshooting

- Check identity, Region and resource ID before changing permissions.
- Resolve every REPLACE placeholder.
- Preserve error output as evidence.
- Escalate cleanup failures immediately.